



MarketingAI

AI-Powered Social Media Manager

Complete Platform Documentation with Screenshots

March 2026 · Next.js · Firebase · Google Gemini AI

TABLE OF CONTENTS

1. Platform Overview & Tech Stack
2. Authentication System
3. Dashboard Home
4. Create Content Page
5. My Posts Page
6. Schedule / Calendar
7. Analytics Page
8. Strategy Agent — Complete Deep Dive
9. Settings Page
10. Backend Architecture & API Routes
11. Adaptive Intelligence — Strategy Evaluation Memory
12. What's Next

1. Platform Overview & Tech Stack

MarketingAI is a full-stack SaaS marketing platform that helps businesses generate AI-written content, schedule it, publish it to Instagram and LinkedIn, analyze its performance, and learn from data using an intelligent Strategy Agent — all in one place.

Core Loop:
Post → Publish → Collect Engagement Data → Analyze → Decide → Post Better

Every piece of the system feeds this loop. The Strategy Agent closes it automatically.

Technology Stack

Layer	Technology	Purpose
Frontend	Next.js 14 (App Router) + TypeScript	Full-stack React framework, routing, SSR
Styling	Tailwind CSS + Framer Motion	UI design + page transition animations
Auth	Firebase Auth + Admin SDK	JWT-based login, token verification per request
Database	Firestore (Firebase)	Posts, users, insights, agent strategies
Primary AI	Google Gemini 2.0 Flash	Captions, sentiment, strategy, replies, tone guard
AI Fallback	Hugging Face Inference API	Backup if Gemini fails
Social APIs	Facebook Graph API v18 + LinkedIn API v2	Publishing, metrics, comments, replies
Images	Pexels API	Free stock photo search via server-side proxy

2. Authentication System

Firebase Auth handles all authentication. Every API route is protected by verifying the user's JWT token before processing any request.

Login Flow — Step by Step

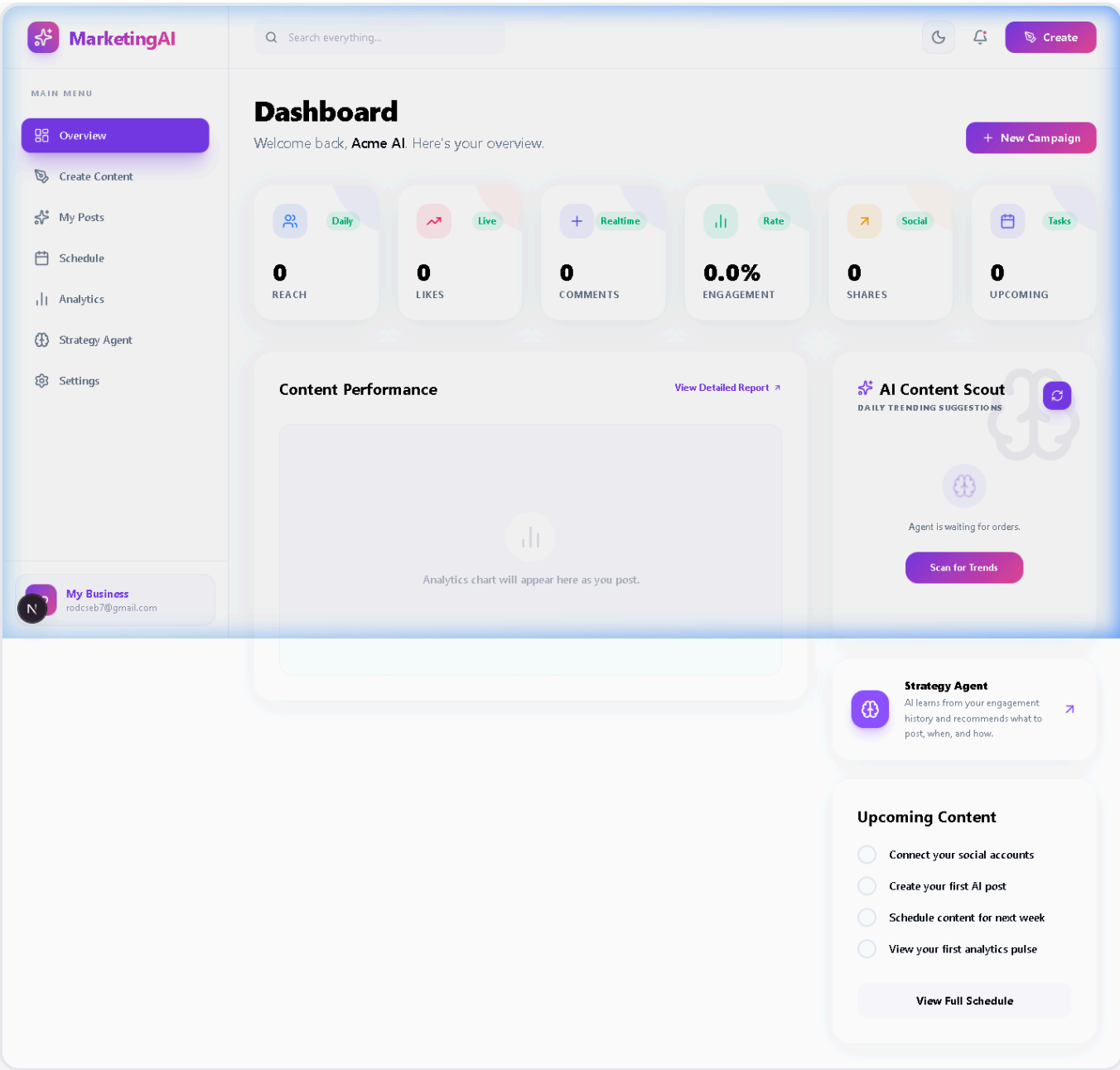
- 1 User enters email + password on `/login`
- 2 Firebase Auth SDK calls `signInWithEmailAndPassword()`
- 3 Firebase issues a short-lived JWT ID Token (1 hour TTL, auto-refreshed)
- 4 Every API call sends: `Authorization: Bearer <token>`
- 5 Each API route calls `getAuth().verifyIdToken(token)` — rejects any expired or tampered tokens

Security Principles

- Tokens stored in React memory, never in `localStorage`
- Dashboard layout auto-redirects to `/login` if no valid session
- Every API verifies `post.userId === uid` before modifying any data (ownership check)
- Social access tokens (Instagram, LinkedIn) stored server-side in Firestore — never exposed to frontend

3. Dashboard Home (`/dashboard`)

The central hub showing a real-time overview of performance, AI suggestions, and quick access to all features.



Dashboard Home — Stats row, AI Content Scout panel, Strategy Agent banner, and Upcoming Content checklist

Page Elements Explained

1 HEADER STATS ROW (6 CARDS)

Card	Data Source	Formula / Logic
Reach	Firestore metrics.reach	Sum of reach across all published posts
Likes	Firestore metrics.likes	Sum of all likes
Comments	Firestore metrics.comments	Sum of all comments
Engagement Rate	Computed on load	(likes + comments) / reach × 100

Shares	Firestore metrics.shares	Sum of all shares
Upcoming	Firestore query	Count of posts with status = "scheduled"

② AI CONTENT SCOUT PANEL

Clicking "Scan for Trends" calls `POST /api/agent/scout`. Gemini generates 3 trending topic posts tailored to the user's business profile and saves them as suggested drafts in Firestore. Posts appear immediately in this panel for review.

③ STRATEGY AGENT BANNER

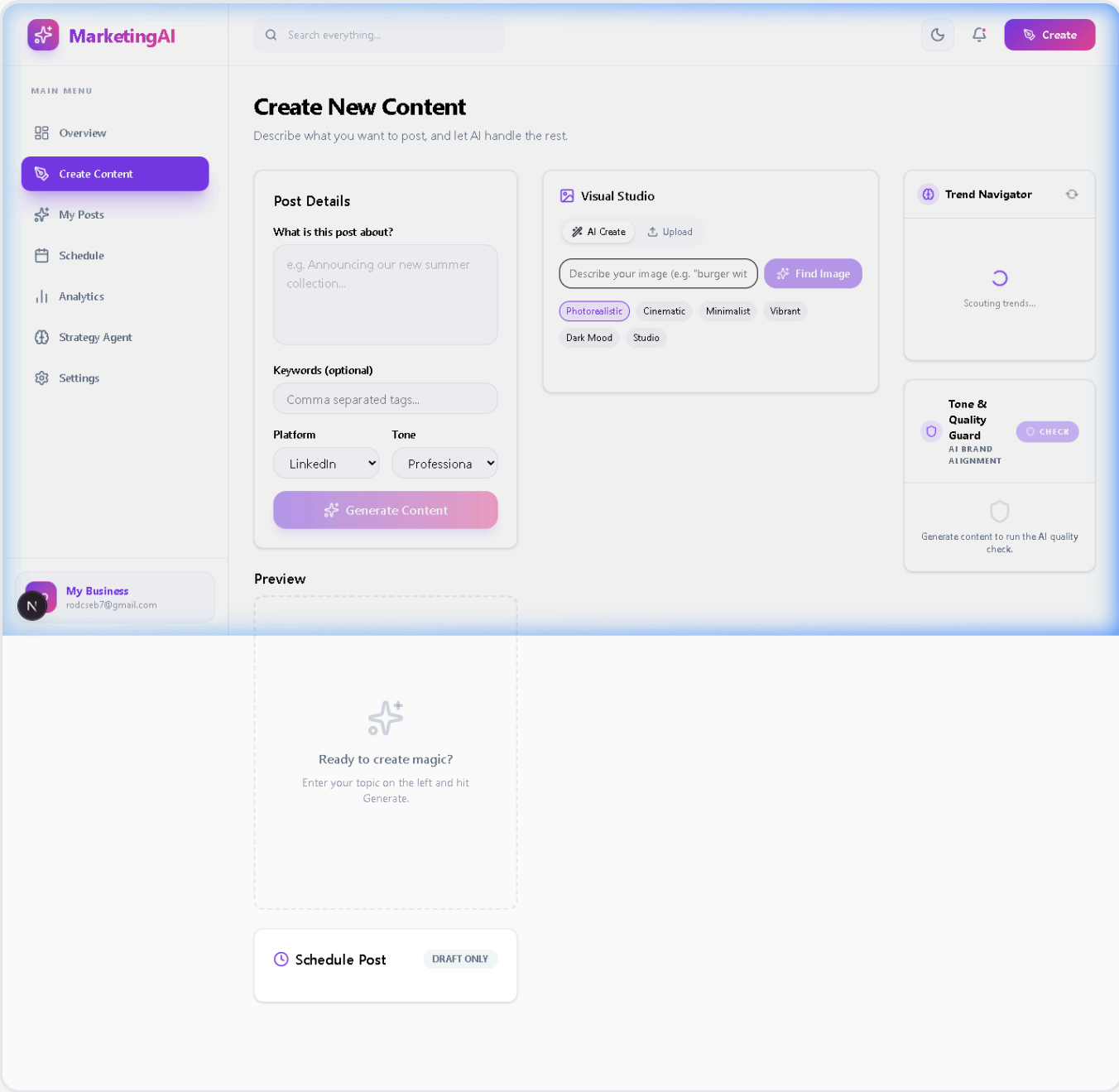
A click-through card that links to `/dashboard/strategy`. Describes the agent's role: it learns from engagement history and recommends what to post, when, and how.

④ UPCOMING CONTENT

Lists upcoming scheduled posts. For new users, shows a "getting started" checklist: connect accounts → create first post → schedule → view analytics.

4. Create Content Page (`/dashboard/create`)

The AI-powered content creation workspace. Enter a topic, pick a tone and platform, and Gemini writes a publish-ready caption with hashtags.



Create Content — Post Details form (left), Visual Studio (center), Trend Navigator + Tone Guard (right)

Left Panel — Post Details Form

Field	Purpose
What is this post about?	Main topic/brief — becomes the core Gemini prompt
Keywords (optional)	Extra terms to weave into the caption
Platform	Instagram / LinkedIn / Twitter — adjusts character limits and writing style
Tone	Professional / Casual / Humorous / Inspirational / Persuasive
Generate Content	Calls POST /api/content/generate → returns Gemini-written caption

Gemini Prompt Structure

```
You are a {tone} social media expert for a {industry} business.  
Write a {platform} post about: {topic}  
Keywords to include: {keywords}  
Include 5 relevant hashtags. Keep it under {character_limit} characters.
```

Center Panel — Visual Studio

- **AI Create** — generates an image using Gemini's image generation capability based on the post topic
- **Find Image** — searches Pexels API (routed through server-side proxy) for free stock photos
- **Upload** — user uploads their own image

Images are stored as CDN URLs (never uploaded to Firebase Storage) to eliminate storage costs.

Right Sidebars

TREND NAVIGATOR

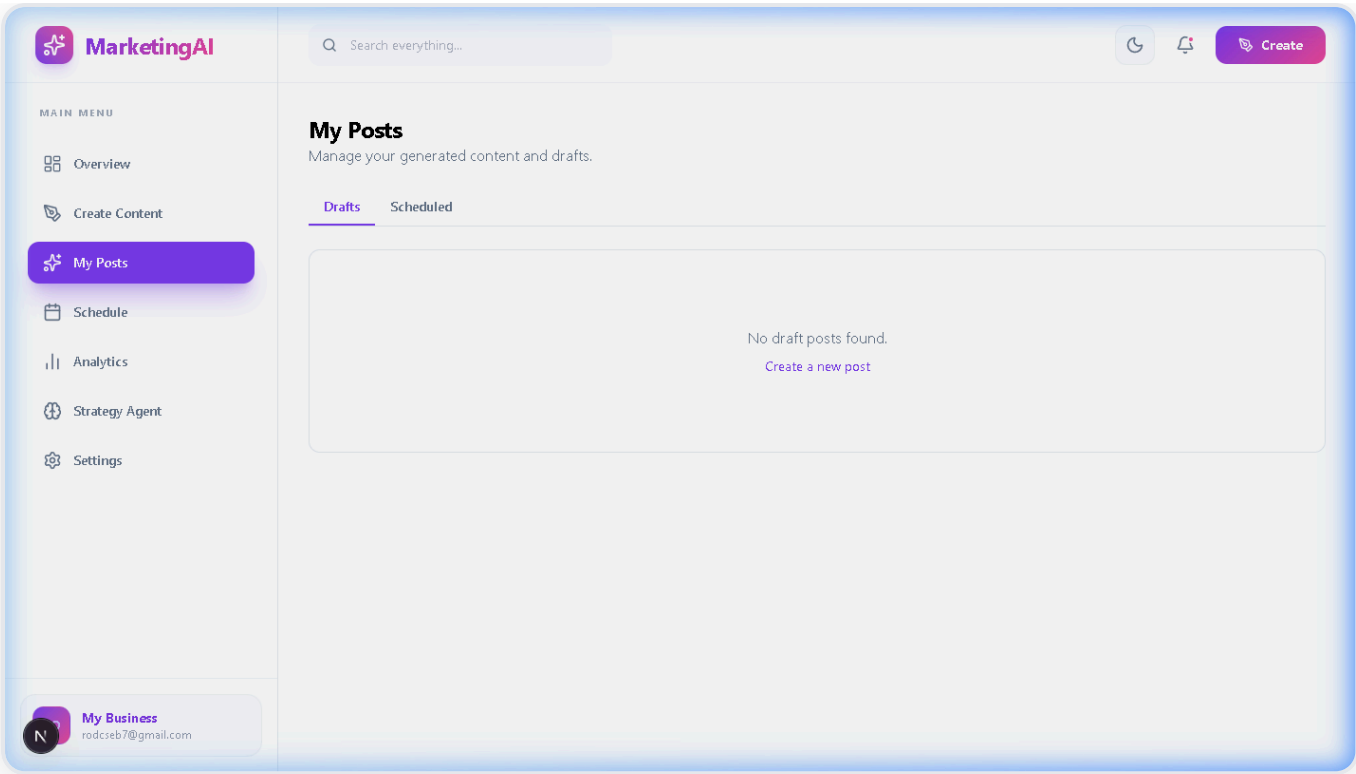
Auto-generates 5 trending topics for the user's industry using Gemini with the current date as context. Refreshes each page load. Click a topic to auto-fill the post form.

TONE & QUALITY GUARD

After generating content, "CHECK" calls POST /api/agent/tone-guard. Gemini compares the writing style to the user's configured brand tone. If misaligned (e.g., too formal for a casual brand), it highlights the issue with a rewrite suggestion.

5. My Posts Page (/dashboard/posts)

Central management hub for all posts regardless of status.



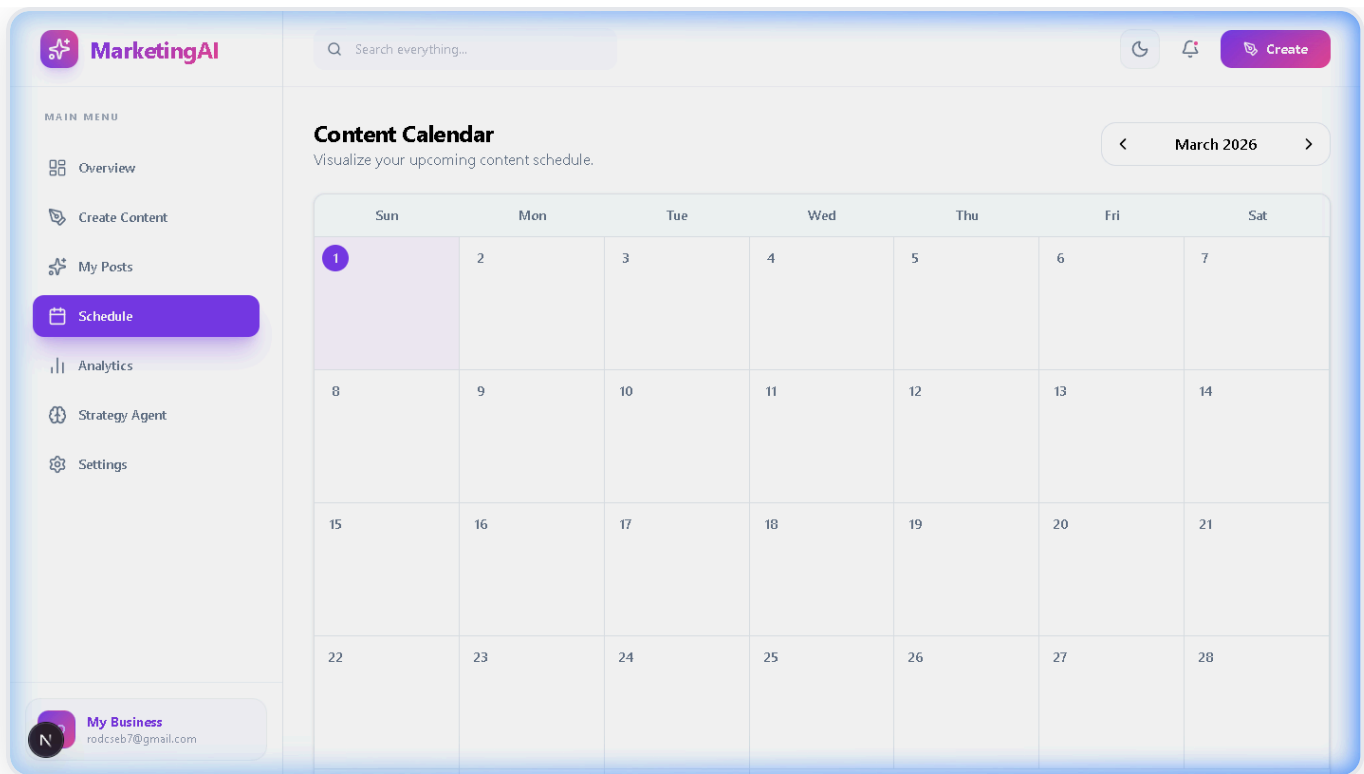
My Posts — All content in a unified view with status badges and quick actions

Post Statuses

Status	Meaning	Available Actions
DRAFT	Saved but not yet scheduled	Schedule, Publish Now, Edit, Delete
SUGGESTED	Created by AI Content Scout	Review, Customize, then Schedule or Publish
SCHEDULED	Queued for future publishing	Edit time, Cancel, Publish Now
PUBLISHED	Live on the platform	View metrics, Analyze Sentiment, Reply to comments

6. Schedule / Calendar (/dashboard/schedule)

Month-at-a-glance view of all scheduled and published content to help visualize posting consistency and gaps.



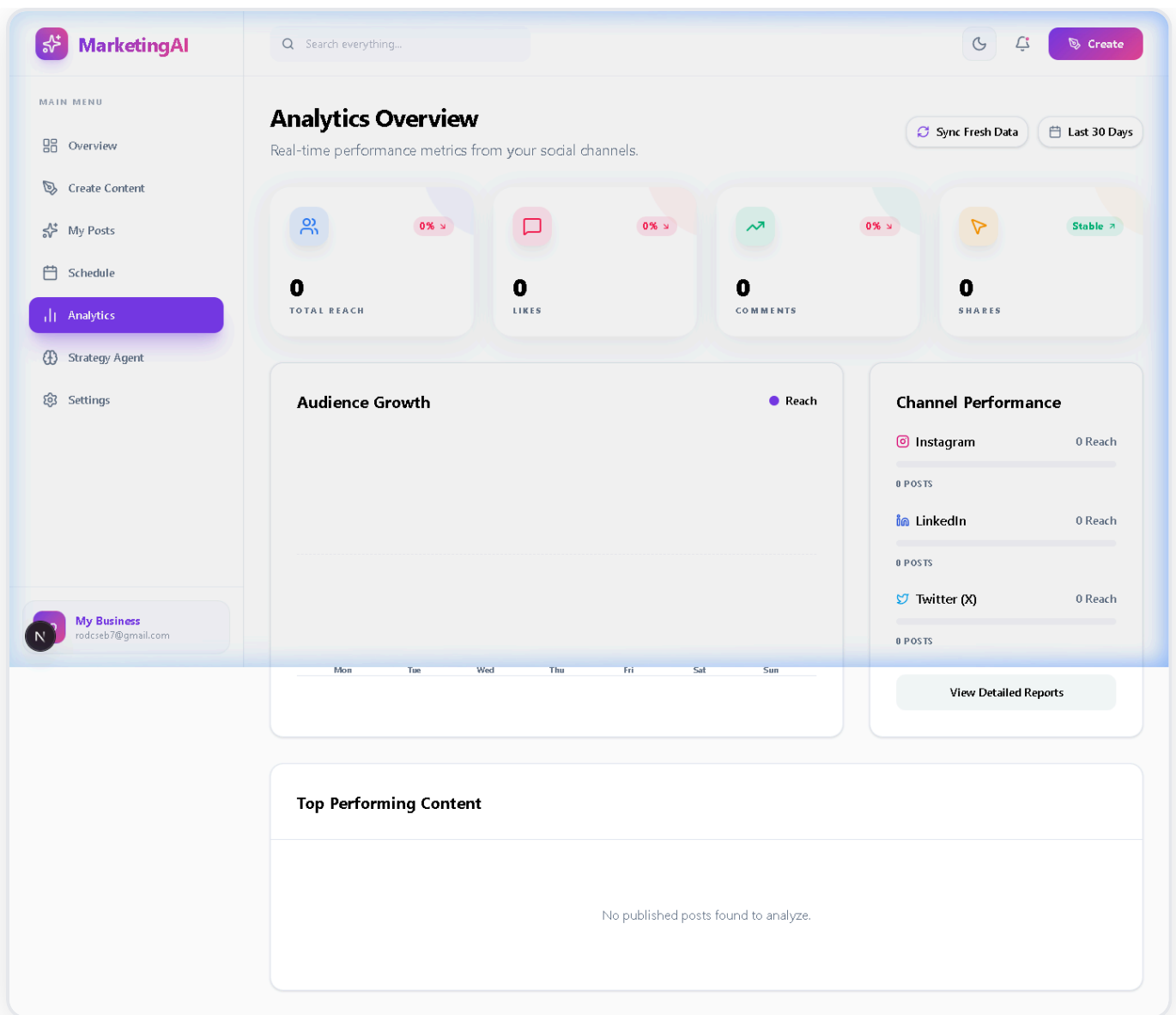
Schedule — Monthly calendar with posts shown as color-coded chips on their scheduled dates

How It Works

- Built with pure React state — no external calendar library
- Each day cell queries Firestore for posts whose `scheduledAt` falls on that date
- Posts appear as colored chips (blue = scheduled, green = published)
- Click any chip to open that post for editing
- Seeing gaps in the calendar motivates consistent posting

7. Analytics Page (/dashboard/analytics)

Full engagement analytics with real-time metrics sync from Instagram and LinkedIn, plus AI-powered sentiment analysis on comments.



Analytics — Total Reach, Likes, Comments, Shares, Audience Growth chart, Channel Performance breakdown

Sync Fresh Data — How It Works

- 1 User clicks "Sync Fresh Data" → calls POST `/api/analytics/sync`
- 2 Fetches all published posts for the user from Firestore
- 3 For each post, calls platform API: Instagram Graph API `/media/{id}/insights` or LinkedIn `/v2/socialMetadata/{id}`
- 4 Writes back metrics: `{ likes, comments, shares, reach }` to each Firestore post document
- 5 **Auto-triggers Observer agent in background** to re-score engagement (fire-and-forget, non-blocking)

Analyze Pulse — Sentiment Modal

Clicking "Analyze Pulse" on any published post opens a modal:

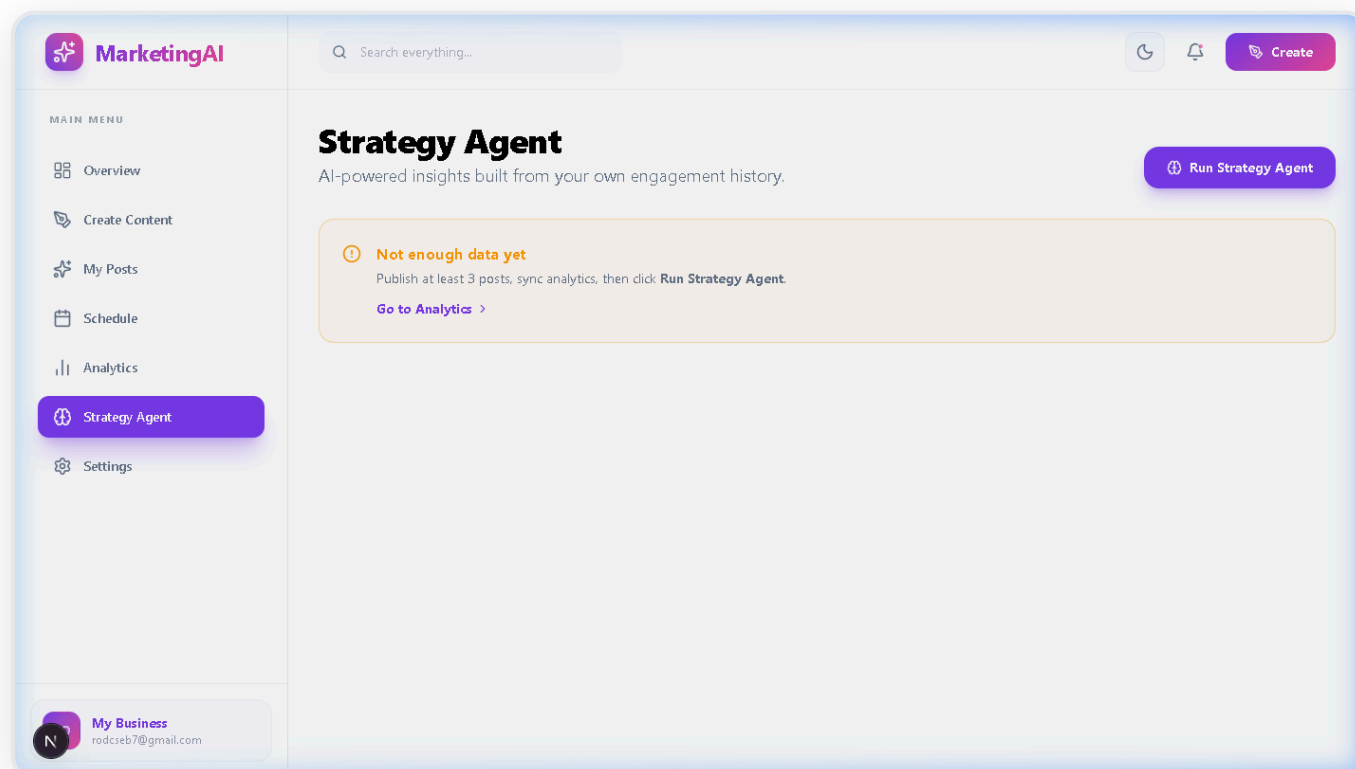
- Fetches all comments from the platform via the comments API
- Sends comment texts to Gemini: *"Analyze sentiment. Return: score 0–100, dominant emotion, top positive/negative themes, summary."*
- Displays: score gauge, positive/negative/neutral split, theme chips, individual comments

AI Reply Suggestions (inside Pulse Modal)

- "Suggest Reply" → POST /api/agent/suggest-reply → Gemini writes brand-aligned reply
- Reply appears in an **editable textarea** — user can modify before posting
- "Post Reply" → POST /api/agent/post-reply → posts directly to Instagram (Graph API /replies) or LinkedIn (nested comments)
- After posting: textarea disabled, shows "Posted!" — prevents double-posting

8. Strategy Agent — Complete Deep Dive (/dashboard/strategy)

The most sophisticated feature. An analytical AI feedback loop that observes what works in your posts, finds patterns, and prescribes a data-driven weekly content strategy.



Strategy Agent — Empty state before first run. Shows the "Run Strategy Agent" button and guidance to publish 3+ posts first.

- Agent runs in 3 sequential stages:
- STAGE 1 — OBSERVE: Score every post with engagement metrics
 - STAGE 2 — ANALYZE: Detect patterns across hours, tones, platforms, categories
 - STAGE 3 — PLAN: Generate a weekly strategy with plain-English reasoning

Stage 1 — OBSERVER MODULE

File: lib/agent/observer.ts | **Trigger:** POST /api/agent/observe | **Auto-runs:** After every analytics sync

ENGAGEMENT SCORE FORMULA

```
engagementScore = (0.5 × likes) + (0.3 × comments) + (0.2 × shares)

Example: 100 likes, 30 comments, 10 shares
= (0.5 × 100) + (0.3 × 30) + (0.2 × 10)
= 50 + 9 + 2 = 61.0
```

Action	Weight	Reason
Likes	0.5	Most common action, moderate signal
Comments	0.3	Higher intent — active engagement is harder to get
Shares	0.2	Strongest virality signal but rarest — lower weight prevents outlier distortion

ENGAGEMENT RATE FORMULA

```
engagementRate = (engagementScore / reach) × 100

Why: Normalizes performance across posts of different audience sizes.
Post A: score 50 from 500 reach = 10% → outperforms
Post B: score 200 from 5000 reach = 4%
```

OTHER FIELDS COMPUTED PER POST

- postHour = publishedDate.getHours() (0–23)
- postDayOfWeek = publishedDate.getDay() (0=Sun, 6=Sat)
- hashtags = all #tags extracted via regex /[#][w]+/g
- contentCategory = promotional / educational / engagement / news (keyword heuristic)

Stage 2 — ANALYZER MODULE

File: lib/agent/analyzer.ts | **Output:** agent_insights/{userId} in Firestore

PATTERN DETECTION — GROUP-BY-AVERAGE

```
For each dimension (hour, tone, platform, category):
  Group all posts by value → calculate mean(engagementScore) per group → find max

Example hourly breakdown:
Hour 07 → avg 12.3 (2 posts)
Hour 12 → avg 18.7 (3 posts)
Hour 19 → avg 41.2 (5 posts) ← bestHour = 19
Hour 21 → avg 22.1 (2 posts)
```

7-DAY TREND DETECTION

```
last7Avg = mean(scores for posts in last 7 days)
prev7Avg = mean(scores for posts 8-14 days ago)
delta = ((last7Avg - prev7Avg) / prev7Avg) × 100

if delta >= 5%: recentTrend = "up" ↑
if delta <= -5%: recentTrend = "down" ↓
else: recentTrend = "stable" →
```

CONFIDENCE SCORE

Posts Analyzed	Confidence	Label
< 3	10%	Very Low — not enough data
3–4	30%	Low
5–9	60%	Medium
10–19	80%	High
20+	100%	Very High

Stage 3 — PLANNER MODULE

File: lib/agent/planner.ts | **Output:** agent_strategy/{userId} in Firestore

WEEKLY POST TARGET

```
confidence >= 80 → weeklyPostTarget = 5
confidence >= 50 → weeklyPostTarget = 4
else → weeklyPostTarget = 3
```

RECOMMENDED POST TIMES

```
Slot 1: bestDay at bestHour:00
Slot 2: bestDay + 2 days at bestHour:00
Slot 3: bestDay + 4 days at bestHour:00

If bestDay=Tuesday, bestHour=19 → ["Tue 19:00", "Thu 19:00", "Sat 19:00"]
```

CONTENT MIX FORMULA

Best Performing Category	Promotional	Educational	Engagement	News
promotional	50%	20%	20%	10%
educational	20%	50%	20%	10%
engagement	20%	20%	50%	10%
news	20%	30%	20%	30%

SAMPLE AGENT REASONING OUTPUT

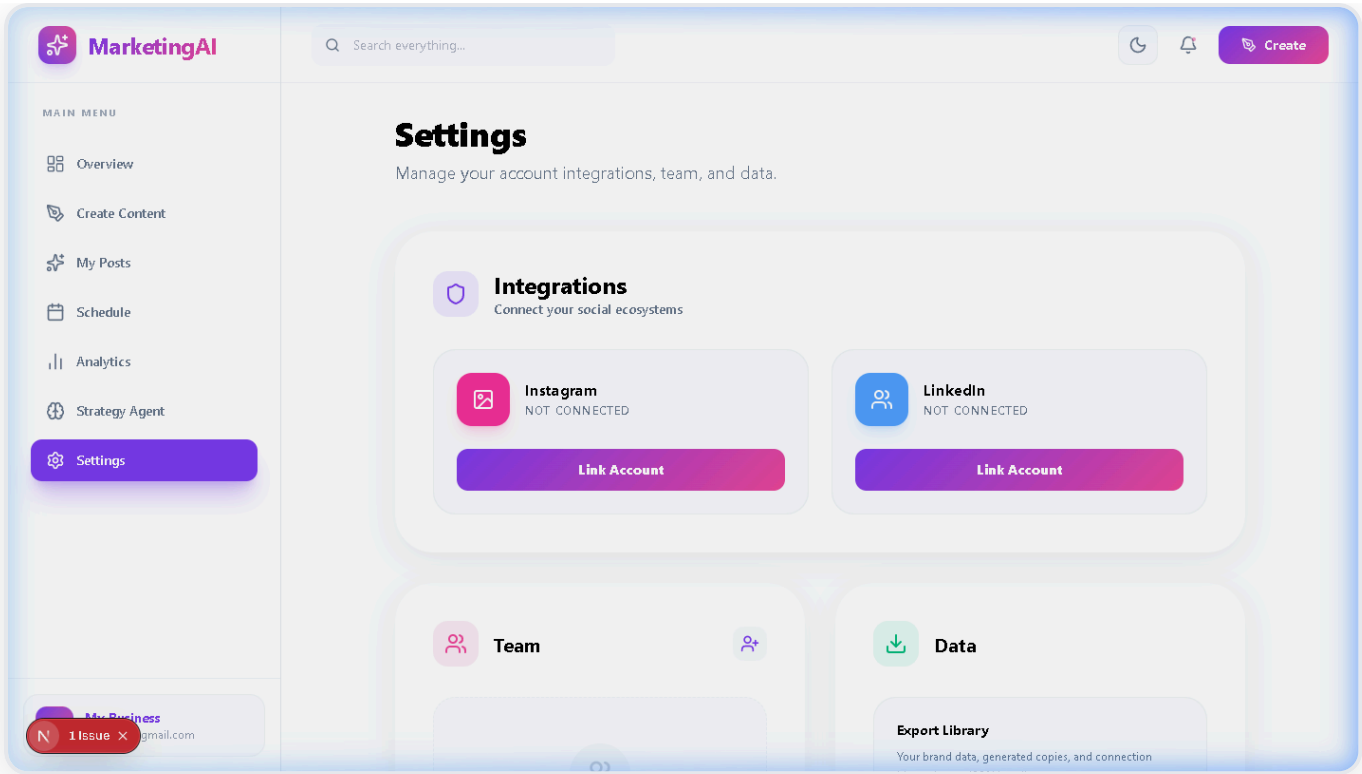
```
"Posts at 19:00 have your highest average engagement score."
"Your 'Casual' tone outperforms others by ~42% in engagement."
"Instagram gives ~120% better engagement than other platforms."
"'promotional' posts drive the most engagement in your feed."
"Your engagement is trending upward — maintain current strategy momentum."
```

Dashboard UI — Every Element Explained

Element	What It Shows	Why It's There
Run Strategy Agent button	Triggers observe → analyze → plan pipeline	Manual control — user decides when to refresh

Posts Analyzed card	Total posts scored by Observer	Shows agent data quality
Avg Engagement Score card	Mean score across all posts	Baseline performance benchmark
Best Platform card	Instagram or LinkedIn — highest avg score	Where to focus effort
Recent Trend card	↑ Up / ↓ Down / → Stable (7-day delta)	Early warning system for performance changes
Hourly Heatmap	24-cell grid, purple opacity = avg score	Instantly shows the best hour to post
Platform Bars	Horizontal score bars per platform	ROI comparison between channels
Tone & Category Bars	Which writing style / content type wins	Informs how to write future posts
Top Hashtags	Tags ranked by avg post score	Tells user which tags drive the most engagement
Weekly Strategy Card	Post target, times, tone, platform, mix, hashtags	Complete action plan for the week
Confidence Badge	High / Medium / Low + %	Prevents over-reliance on sparse data
Agent Reasoning list	Plain-English explanation for every recommendation	Builds trust — every suggestion has a data reason

9. Settings Page (/dashboard/settings)



Settings — Integrations tab showing Instagram and LinkedIn connection cards with "Link Account" OAuth buttons

Instagram Connection — OAuth Flow

- 1 "Link Account" → redirects to Meta OAuth via GET /api/auth/instagram
- 2 User approves on Meta consent screen (permissions: instagram_manage_comments, instagram_content_publish)
- 3 Meta redirects to /api/auth/instagram/callback with an authorization code
- 4 Server exchanges code for long-lived access token and saves to Firestore: users/{uid}/connections/instagram

LinkedIn Connection

Same OAuth flow via GET /api/auth/linkedin. Stores { linkedinId, accessToken, memberId } in Firestore.

10. Backend Architecture & API Routes

Method	Route	Description
GET/POST	/api/user/profile	Fetch or update business profile

GET/POST	/api/posts	List all posts / create new post
GET/PATCH/DELETE	/api/posts/{id}	Read / update / delete a single post
GET	/api/posts/{id}/comments	Fetch comments from the social platform
POST	/api/content/generate	Generate AI caption via Gemini
GET	/api/auth/instagram	Start Instagram OAuth redirect
GET	/api/auth/instagram/callback	Complete token exchange + save
GET	/api/auth/linkedin	Start LinkedIn OAuth redirect
GET	/api/auth/linkedin/callback	Complete LinkedIn token exchange
POST	/api/analytics/sync	Sync metrics from platforms + auto-trigger Observer
POST	/api/analytics/sentiment	Run Gemini sentiment analysis on comments
GET	/api/analytics/timing	Best posting time analysis
POST	/api/agent/observe	Run Observer module manually
POST	/api/agent/analyze	Run Analyzer → write agent_insights
GET/POST	/api/agent/plan	Fetch existing / generate new weekly strategy
POST	/api/agent/scout	Run Content Scout — generate suggested posts
POST	/api/agent/suggest-reply	Generate AI reply to a comment
POST	/api/agent/post-reply	Post a reply directly to the platform
POST	/api/agent/tone-guard	Validate content against brand tone
POST	/api/agent/evaluate	Trigger weekly evaluation → compute delta → update weights
GET	/api/agent/weights	Fetch current adaptive weights + recent performance logs
POST	/api/agent/weights	Reset adaptive weights to 1.0 baseline

11. Adaptive Intelligence — Strategy Evaluation Memory

A **closed-loop evaluation engine** that learns from every weekly strategy, adjusts internal weights based on real performance, and continuously improves future recommendations automatically.

Closed Loop:
Planner predicts score → Posts execute → Evaluator measures actual score
→ Delta computed → Adaptive weights updated → Next strategy is smarter

Firestore Collections Added

Collection	Key Fields	Purpose
agent_strategies	strategyId, userId, predictedCompositeScore, status, appliedWeights	Stores each generated weekly strategy with its predicted score
strategy_performance_logs	strategyId, actualCompositeScore, delta, postsCount, consistencyScore	Records actual vs. predicted performance after evaluation
adaptive_weights	userId, bestHourWeight, toneWeight, platformWeight, categoryWeight, updateCount, consecutiveUnderperformance	Per-user learning state persisted across all runs
follower_snapshots	userId, followerCount, recordedAt	Weekly follower counts used to compute growth rate

Composite Score Formula

```
compositeScore =  
(0.40 × normalizedEngagementScore)  
+ (0.30 × normalizedEngagementRate)  
+ (0.20 × normalizedGrowthRate)  
+ (0.10 × consistencyScore)
```

All components normalized to 0-100 range.

Adaptive Weight Rules

Delta (actual – predicted)	Weight Change
delta ≥ +10 (overperformed)	+0.1 boost (capped at 1.5×)
delta ≤ –10 (underperformed)	–0.1 reduction (floored at 0.5×)
–10 < delta < +10	No change (stable)
2 consecutive underperformances	Safety reset → all weights back to 1.0

Files Added / Updated

File	Role
lib/agent/scoring.ts	Composite score formula & normalization functions
lib/agent/weights.ts	Fetch, update, clamp, and reset adaptive weights in Firestore
lib/agent/evaluator.ts	Weekly evaluation orchestrator — closes the loop
lib/agent/planner.ts	UPDATED — applies weights, saves predictedCompositeScore
app/api/agent/evaluate/route.ts	POST endpoint to trigger evaluation
app/api/agent/weights/route.ts	GET weights + logs · POST reset weights
app/dashboard/strategy/page.tsx	UPDATED — Adaptive Intelligence UI panel added

UI Panel Features

- **Adaptive Weights bars** — Best Hour, Tone, Platform, Category (green=boosted, red=reduced, blue=neutral)
- **Run Weekly Evaluation button** — triggers full loop, shows delta + update status
- **Performance History table** — Date · Predicted · Actual · Delta (↑/↓) · Posts · Consistency %
- **Reset Weights button** — manual override back to 1.0 baseline
- **Warning banner** — shows consecutive underperformance count approaching safety reset

End-to-End Flow

- 1 User clicks **Run Strategy Agent** → Planner fetches weights → generates strategy with weighted preferences → saves predictedCompositeScore
- 2 User publishes posts throughout the week (existing flow)
- 3 User clicks **Run Weekly Evaluation** → Evaluator fetches posts → calculates actualCompositeScore
- 4 $\Delta = \text{actual} - \text{predicted}$ → if $|\Delta| \geq 10$, weights auto-adjust
- 5 Next **Run Strategy Agent** uses updated weights → smarter recommendations

12. What's Next

Phase 7 — Team Workspaces (Not Yet Built)

Feature	What It Enables
Workspace Firestore collection	Group users + posts under one workspace
Membership system	Link multiple users to the same workspace
Workspace-scoped posts	Posts belong to workspace, not individual user
Invitation by email	Invite team members with a magic link
Role-Based Access Control	Admin: publish Member: draft only

Optional Future Enhancements

Feature	Value
Strategy Executor	Agent auto-schedules posts based on the strategy plan
Twitter/X Integration	Third platform connector
Engagement Prediction	Score a draft before publishing using historical patterns
Monthly Campaign Planning	Full month strategy, not just one week

Push Notifications

Alert when posts perform unusually well or drop

PDF Export

Download performance reports as formatted PDFs

MarketingAI Platform Documentation — Updated March 2026 v2.0

Built with Next.js · Firebase · Google Gemini AI · Instagram Graph API · LinkedIn API